

Isaac Legene

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EDUCATION

Georgia Institute of Technology

M.S. in Robotics - Electrical and Computer Engineering (GPA: 4.0)

Atlanta, GA

2025 – 2026

Missouri University of Science and Technology

B.S. in Engineering Management - Industrial Engineering (GPA: 3.51)

Rolla, MO

2022 – 2024

ROBOTICS RESEARCH

Georgia Institute of Technology: LIDAR Lab

Graduate Researcher - Humanoid Loco-Manipulation

Atlanta, GA

2025 – 2026

Diffusion Policy Fine-tuning for Humanoid Loco-Manipulation via Reinforcement Learning

IEEE RA-L 2026

Tactile-Based Whole-Body Manipulation via Force-Supervised Contact-Aware Planning

IEEE RA-L 2026

ROBOTICS AND AUTOMATION EXPERIENCE

Tesla

Automation Controls Engineering Intern

Material Flow Automation - autonomous mobile robotics

Austin, TX

May 2026 – Aug 2026

- Support autonomous mobile robot fleet validation and deployment in a manufacturing environment.
- Repaired and debugged AMR, PLC, and LiDAR scanner issues to reduce downtime and support production operations.
- Developed a script to help push configuration updates across the AMR fleet.

Rivian

Automation Controls Engineering Intern

Robotics simulation and virtual commissioning R2

Normal, IL

Jan 2025 – Aug 2025

- Supported simulation and virtual commissioning design reviews for R2 Body-in-White.
- Developed full-scope production line simulations for upcoming R2 builds.
- Assisted the development of a UI add-in tool that accelerated task timelines in simulation.
- Tested Unity for virtual commissioning as a lower-cost replacement for traditional simulation and VC.

Tesla

Automation Controls Engineering Intern

Body-in-White Production Equipment Integration for Models 3 & Y

Fremont, CA

Jan 2024 – Aug 2024

- Oversaw and supported contract work and inventory for Body-in-White production equipment integration.
- Supported the integration of new production controls and processes to ramp up production.
- Designed and implemented HMIs throughout the body shop for future troubleshooting needs.
- Implemented devices and developed PLC programs that were integrated into the production process.
- Completed training in Fanuc/KUKA (101–301), AC/DC circuits, multimeter usage, pneumatics, and PLC programming.

Dakota Integrated Systems

Automation Controls Engineering Intern

Tier 1 supplier for the Ford Expedition and Lincoln Navigator

Chicago, IL

May 2023 – Aug 2023

- Assisted sustainability practices for all process automation systems to ensure quality and continuity of design.
- Troubleshoot and maintain network performance, connectivity, and hardware to mitigate future failures.
- Developed and updated standard work instructions for all manufacturing assembly lines and stations.

Roeslein & Associates

Electrical & Instrument Designer Intern

Modular fabrication of biogas collection systems on farms domestically

Saint Louis, MO

May 2022 – Aug 2022

- Reviewed PLC programs for instrumentation and automated procurement processes for biogas systems.
- Coordinated with vendors and managed purchase orders for upcoming projects to mitigate downtime.
- Developed instrumentation datasheets used in P&ID documents and inventory tracking.

TECHNICAL SKILLS

Software: Git, Process Simulate, Isaac Lab, ROS, Unity, Ignition, Linux, MuJoCo, Docker

PLC and Robot Platforms: Beckhoff, Allen-Bradley, Siemens, Kuka, Fanuc, Kawasaki

Languages: C#/.NET, C++, Python

Key Courses: Computer Vision, Deep Reinforcement Learning, Machine Learning